

PROJECT DELIVERABLE - PHASE 2

Project Design Phase

System Architecture, Data Flow, UI/UX Layout & Database Schema

Platform: ServiceNow PDI dev423661 | Global Scope | September 2026

1. System Architecture Design

The Metro Ticket Generating System employs a modern 3-Tier Enterprise Cloud Architecture hosted on the ServiceNow platform, integrating Commuter Portal, Application Logic, and Relational Data tiers:

- Commuter Presentation Tier: ServiceNow Service Portal (/sp), responsive Bootstrap/AngularJS catalog view, spModal dialog.
- Application Logic Tier: Service Catalog Engine, onChange Client Scripts, onSubmit QR generation, Catalog UI Policies.
- External Integration Tier: RESTful QR Server API (api.qrserver.com) generating dynamic 150x150 scannable QR ticket matrices.
- Enterprise Data Tier: Master station table (u_metro_station_details), Catalog request tables (sc_request, sc_req_item).

2. Data Flow Architecture (DFD Level 0 & Level 1)

The data flow traverses through station selection, fare calculation, UI policy evaluation, QR generation, and order persistence:

Step	Source Component	Destination Component	Data Transferred / Action
1	Commuter Browser	Service Portal (/sp)	User searches and opens 'Book A Metro Tic..
2	Service Catalog Form	u_metro_station_deta..	Reference queries retrieve valid stations..
3	Passenger Selector	onChange Client Script	Passes starting_from, going_to, type_of_j..
4	Fare Engine	GlideForm Amount Field	Populates amount_for_single_journey or am..
5	Payment Mode Selec..	Catalog UI Policy	Evaluates mode_of_payment == 'Others'; re..
6	Order Now Click	onSubmit Client Script	Intercepts submit, constructs QR URL with..
7	onSubmit Script	spModal / metro_qr_w..	Opens popup modal displaying dynamic QR t..
8	Modal OK Click	ServiceNow DB Engine	Creates persistent Request (sc_request) &..

3. Database Schema Design (Custom Master Table)

A dedicated custom table was architected in ServiceNow System Definition to maintain station master data, referenced by catalog items:

Column Label	Column Name	Data Type	Length	Display Field	Purpose
Station Name	u_station_name	String	40	True	Human-readable metro stati..
Sys ID	sys_id	GUID	32	False	ServiceNow 32-character un..
Created	sys_created_on	Date/Time	-	False	Audit timestamp recording ..
Created By	sys_created_by	String	40	False	Username of administrator ..
Updated	sys_updated_on	Date/Time	-	False	Audit timestamp of latest ..
Updates	sys_mod_count	Integer	-	False	Incremental revision count..

4. Service Catalog Variable Specifications

The booking form comprises 8 precisely typed Catalog Variables configured under Maintain Items:

Variable Name	Display Label	Type	Mandatory Read Only Behavior / Rules		
starting_from	Starting From	Reference	Yes	No	References u_metro_s..
going_to	Going to	Reference	Yes	No	References u_metro_s..
type_of_journey	Type of journey	Multiple Ch..	Yes	No	Radio buttons: Singl..
no_of_passengers	No of Passengers	Select Box	Yes	No	Dropdown: 1, 2, 3, 4..
amount_for_single_..	Amount for single..	Single Line..	No	Yes	Auto-calculated fare..
amount_including_r..	Amount including ..	Single Line..	No	Yes	Auto-calculated fare..
mode_of_payment	Mode of Payment	Select Box	Yes	No	Options: UPI, Card, ..
enter_payment_mode	Enter payment mode	Single Line..	Cond.	No	Revealed & mandatory..

5. UI/UX Wireframe & Widget Design

The interface consists of two synchronized viewports designed for high clarity and error prevention:

- Service Catalog Form View: Standard ServiceNow 2-column responsive layout with station pickers, passenger chips, and locked fare fields.
- Metro QR Widget (metro_qr_widget): AngularJS template modal window (spModal.open) displaying a centered QR code image with instructions.

6. Security & Access Control Design

Access Control Lists (ACLs)

ServiceNow generated four default table-level ACL rules for u_metro_station_details (read, create, write, delete). Standard authenticated commuters possess read access to query station lists, while write/create/delete are restricted to System Administrator.